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Research

Mulligan manual therapy added to exercise improves headache frequency, intensity and disability more than exercise alone in people with cervicogenic headache: a randomised trial

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KEY WORDS

Cervicogenic headache
Exercise
Manual therapy
Mobilisation with movement
Placebo effect

ABSTRACT

Question: What is the effect of a 4-week regimen of Mulligan manual therapy (MMT) plus exercise compared with exercise alone for managing cervicogenic headache? Is MMT plus exercise more effective than sham MMT plus exercise? Are any benefits maintained at 26 weeks of follow-up? **Design**: A three-armed, parallel-group, randomised clinical trial with concealed allocation, blinded assessment of some outcomes and intention-to-treat analysis. **Participants**: Ninety-nine people with cervicogenic headache as per International Classification of Headache Disorders (ICHD-3). **Interventions**: Participants were randomly allocated to 4 weeks of: MMT with exercise, sham MMT with exercise or exercise alone. **Outcome measures**: The primary outcome was headache frequency. Secondary outcomes were headache intensity, headache duration, medication intake, headache-related disability, upper cervical rotation range of motion, pressure pain thresholds and patient satisfaction. Outcome measures were collected at baseline and at 4, 13 and 26 weeks. **Results**: MMT plus exercise reduced headache frequency more than exercise alone immediately after the intervention (MD between groups in change from baseline: 2 days/month, 95% CI 2 to 3) and this effect was still evident at 26 weeks (MD 4 days, 95% CI 3 to 4). There were also benefits across all time points in several secondary outcomes: headache intensity, headache duration, headache-related disability, upper cervical rotation and patient satisfaction. Pressure pain thresholds showed benefits at all time points at the zygapophyseal joint and suboccipital areas but not at the upper trapezius. The outcomes in the sham MMT with exercise group were very similar to those of the exercise alone group. **Conclusions**: In people with cervicogenic headache, adding MMT to exercise improved: headache frequency, intensity and duration; headache-related disability; upper cervical rotation; and patient satisfaction. These benefits were not due to placebo effects. **Trial registration**: CTRI/2019/06/019506. **[Satpute K, Bedekar N, Hall T (2024) Mulligan manual therapy added to exercise improves headache frequency, intensity and disability more than exercise alone in people with cervicogenic headache: a randomised trial. *Journal of Physiotherapy* 70:224–233]**

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Introduction

Headache disorders are the second leading cause of years lived with disability globally.[1,2] The direct and indirect burden of headache pose high financial costs to a person and society.[3,4] One sub-category of headache is cervicogenic headache (CGH), which is a secondary headache according to the third edition of the International Classification of Headache Disorders (ICHD-3)[3] attributed to cervical musculoskeletal dysfunction.[4] The prevalence of CGH among the general population ranges between 4 and 7%.[5,6]

The diagnosis of CGH can be made using criteria presented by the International Headache Society[3] or criteria developed by the International Study Group.[4] CGH is a non-throbbing, unilateral, side-locked headache that originates in the cervical spine and gradually spreads to the occipital, temporal and orbital areas.[7] It is associated with neck pain or stiffness and is often aggravated by sustained neck postures, repeated neck movements or physical activity.

A recent systematic review identified various musculoskeletal impairments that are present in people with CGH and not in headache-free controls,[8] including: reduced cervical and upper cervical range of motion (ROM); reduced strength and endurance of cervical flexor and extensor muscles; and impaired motor control of the deep neck flexor muscles.[8] A cluster of cervical musculoskeletal impairments might assist in confirming a cervical disorder as a cause of the headache.[9] The combination of restricted movement, palpable upper cervical joint dysfunction and impairment in the cranio-cervical flexion test has excellent sensitivity and specificity for identifying CGH.[10] Clinically, CGH needs to be differentiated from

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migraine and tension-type headaches due to the partial overlap of symptoms between them.[11–13] The dominant impairments differentiating CGH from migraine are restriction of upper cervical ROM on the flexion-rotation test by 18 deg and reduction in cervical flexor muscle strength.[14] Furthermore, headache experts agree on 11 outcome measures (including the flexion rotation test, assessment of trigger points and manual joint palpation) that should be considered for assessment of musculoskeletal impairments.[15] These are in addition to the assessment of disability[16] and headache parameters such as headache frequency and duration, which are important to classify the headache type and its recovery.[3]

Physiotherapy management including exercise and manual therapy has been used in the management of CGH.[17,18] Exercise programs in isolation are reported to be as effective for the management of CGH[19,20] and have also been used as a comparator intervention against manual therapy.[19] Exercise programs for CGH include specific motor control training of the deep neck flexors or comprise a combination of strength, endurance and mobility training of the upper quadrant muscles.[19–21] Systematic reviews have identified benefit from manual therapy for CGH;[22–25] that evidence is further supported by a consensus document by headache experts from different countries.[26] The benefits of manual therapy for CGH have been demonstrated primarily in the short term, with effect sizes varying from small to moderate.[17] The evidence is unclear for long-term effectiveness.[23,27] Patients commonly favour manual therapy because they believe it will reduce symptom severity.[28]

There have been a number of studies evaluating the effects of Mulligan manual therapy (MMT) in the management of CGH.[29–32] It comprises various techniques that use pain-free, sustained, joint mobilisation forces in precise directions applied to the upper cervical spine in a sequential manner.[33] Each MMT technique is only recommended to be added to the treatment regimen if substantial improvement is observed in headache symptoms or cervical impairments at the time of application.[33] The hypoalgesic mechanism of MMT is similar to other manual therapy techniques, wherein it is suggested to alter peripheral (input), central (processing) and response (output) mechanisms of nociception.[33] MMT has been shown to have promising results in the management of CGH in the short term.[34,35] Furthermore, MMT induced a long-lasting and clinically important improvement in upper cervical ROM in one randomised trial, with long-term reduction in headache severity in people with CGH.[29] However, many of these studies have evaluated only a single technique in isolation[29,30,32,36] and not as a comprehensive approach using multiple techniques as recommended in the application of MMT.[33,37] In addition, the evidence for the effects of MMT on headache frequency is unclear. Thus, this study hypothesised that MMT will show additional benefits over exercise and sham MMT on headache frequency, intensity and disability over the short and long term in the management of patients with CGH.

Therefore, the research questions for this randomised trial were:

1. What is the effect of a 4-week regimen of MMT plus exercise compared with exercise alone for managing CGH?
2. Is MMT plus exercise more effective than sham MMT plus exercise?
3. Are any benefits maintained at 26 weeks of follow-up?

Methods

Design

A pragmatic, randomised controlled trial was conducted with concealed allocation, assessor blinding of objective measures compared between the MMT plus exercise group and the sham MMT plus exercise group, and intention-to-treat analysis. People with headache who were referred to the physiotherapy outpatient department were consecutively screened and, where eligible, invited to join this trial. Demographic data collected at baseline included age, sex, symptom duration, occupation, height and weight. Eligible and consenting participants with CGH were randomly allocated into one of three parallel groups: MMT plus exercise, sham MMT plus exercise and exercise alone. The random allocation list was computer-generated and concealed in sequentially numbered, sealed, opaque envelopes. The interventions were applied for 4 weeks and participants were followed up for 26 weeks after randomisation. The primary outcome was headache frequency and the secondary outcomes included other headache measures, medication use, pressure pain thresholds, cervical ROM and patient satisfaction. The study protocol was registered and published prospectively.[38] The study is reported according to the recommendations of the Consolidated Standards of Reporting Trials (CONSORT) statement.[39]

Participants

A diagnosis of CGH was made by the primary investigator under the guidance of a medical practitioner as per the ICHD-3 (Box 1).[3] Additional inclusion criteria were: age $>$ 18 years and $<$ 60 years; headache intensity $>$ 6 on a 10-cm visual analogue scale at enrolment; $\geq$ 1-year history of headache with a mean frequency of $\geq$ 1/week; hypomobility of the upper cervical spine (C0 to C3) on manual examination; and reproduction of the headache on palpation of the upper cervical spine (C0 to C3). Participants were excluded if clinical testing revealed: instability of the upper cervical spine; evidence of cervical arterial insufficiency; history of vertigo or dizziness; rheumatoid arthritis, ankylosing spondylosis or cervical spine fractures; pregnancy; cognitive compromise; migraine; or any other contraindications to manual therapy.[38]

Interventions

Participants in the exercise alone group received a structured exercise program comprising cervical flexion loading exercises,[40] scapular retraction in a prone position, passive static self-stretching exercises and active mobility exercises of the neck. The exercise component took 20 minutes/session. The MMT with exercises group pragmatically received additional MMT, which took an additional 10 minutes/session. During the initial consultation, a trial headache sustained natural apophyseal glide (SNAG), reverse headache SNAG, modified headache SNAG and upper cervical traction technique were trialled in sequence to determine which technique, if any, would reduce headache intensity. The next technique in sequence was only applied if no reduction in headache intensity was observed with the application of the first technique. If the headache intensity reduced with a technique, it was repeated as the treatment. At subsequent consultation, the same technique was used if the subject presented with headache, otherwise a C1–C2 rotation SNAG was used to improve upper cervical rotation ROM if found to be restricted. During

Box 1. Cervicogenic headache diagnostic criteria of the International Classification of Headache Disorders (ICHD-3).[3]

A. Any headache fulfilling criterion C
B. Clinical and/or imaging evidence of a disorder or lesion within the cervical spine or soft tissues of the neck, known to be able to cause headache
C. Evidence of causation demonstrated by at least two of the following:
   a. Headache has developed in the temporal region in relation to the onset of the cervical disorder or appearance of the lesion
   b. Headache has significantly improved or resolved in parallel with the improvement in or the resolution of the cervical disorder or lesion
   c. Cervical range of motion is reduced and headache is made significantly worse by provocative manoeuvres applied to the neck
D. Not better accounted for by another ICHD-3 diagnosis

application of the MMT techniques, minor alteration in the direction of force was performed to achieve complete reduction of headache symptoms wherever applicable. The sham MMT plus exercise group received the additional sham MMT intervention,[38] replicating the headache SNAG technique but without any manual force applied; the position was held for 10 to 30 seconds with six repetitions. The duration of the sham treatment session was similar to that in the MMT group. Participants in the MMT and sham MMT groups continued to receive the same exercise protocol as the exercise only group. Each participant received six treatment sessions as per group allocation spread over 4 consecutive weeks. Two treatment sessions were provided each week for the first 2 weeks followed by one treatment session per week for the following 2 weeks. Each treatment session was for a maximum of 30 minutes. The principal investigator had training in manual therapy and > 15 years of clinical experience in using the interventions. The principal investigator avoided any communication with the patient, except to gauge the current headache intensity during the application of treatment techniques. All participants were requested to perform the home exercise program once a day commencing at the end of the 4-week intervention period. The type of analgesics prescribed by a medical practitioner was not altered during the study period but the subjects were allowed to adjust the dose of analgesics as per their need.

Outcome measures

Primary outcome

The primary outcome measure was headache frequency measured as headache days/month. At baseline, subjects were asked to estimate the number of days with headache for the previous month.

Secondary outcomes

The secondary outcome measures included headache intensity measured from 0 (no pain) to 10 (worst imaginable pain), headache duration measured in hours/week and medication intake measured as tablets/week. A headache diary was used to record headache-related parameters, which were extracted at reassessment. Headache-related disability was recorded with the Headache Activities of Daily Living Index (0 to 45).[41] Patient satisfaction score was assessed using a 0-to-100 numerical rating scale. Upper cervical rotation ROM was measured in degrees to both sides with the flexion-rotation test. The other secondary outcomes were the pressure pain threshold measured over the suboccipital muscle area, C2–C3 zygapophyseal joint area, upper trapezius muscle area bilaterally and the tibialis anterior muscle area. An assessor who was blind to group allocation collected outcome measures at baseline after enrolment and at 4, 13 and 26 weeks following treatment commencement. To gauge the success of blinding, participants in the MMT with exercise and sham MMT with exercise groups also indicated their agreement with the statement 'While undertaking physiotherapy management, I received a manual therapy procedure'.

Data analysis

The sample size was calculated based on the primary outcome of headache frequency. According to the International Headache Society guidelines (2018),[3] a 50% reduction in headache frequency is considered to be a clinically worthwhile effect. We anticipated that the participants would have a headache frequency of 4 days/month, so a 50% reduction would equate to a between-group difference of 2 days/month. A sample size calculation with power of 0.8, an alpha of 0.05, 2 days/month as the effect size sought, and an assumed standard deviation of 2.4 indicated 24 participants per group. To account for the multiple comparisons in this three-arm trial and to allow for 10% loss to follow-up, this was increased to 33/group for a total of 99 participants.

Statistical analysis was performed using commercial software[a]. Inferential statistics were conducted according to the principle of intention to treat. For all analyses, statistical tests were two-tailed. The distribution of each outcome dataset was assessed with the Shapiro-Francia test. For outcomes with normally distributed data, means and standard deviations were calculated for each variable. A repeated-measures, general linear model (independent factor was group exercise alone versus MMT with exercise versus sham MMT with exercise and repeated factor was time: baseline, post-intervention, and at 13 and 26 weeks of follow-up) was used to evaluate the differences in all outcome variables. The results are presented as the mean between-group difference (MD) and 95% confidence interval (95% CI) in change from baseline to post-intervention and to first and second follow-up (13 and 26 weeks) for all outcomes. For other outcomes, medians and interquartile ranges were calculated for each outcome and median difference in change from baseline and 95% CI were calculated using Hodges-Lehmann estimation.

The smallest worthwhile effect is the smallest between-group difference in an outcome that patients typically believe outweighs the cost, risk, effort and inconvenience of undertaking a given intervention.[42] For physiotherapy interventions, this effect has been estimated by patients in a benefit-harm trade-off study as a 20% improvement for pain and disability across a range of pain sites.[43]

Results

Compliance with the study protocol

There were no deviations from the study protocol. All participants received their allocated six intervention sessions. No participants in any group reported receiving other interventions during the study. No adverse events were reported following any intervention.

Flow of participants through the study

Ninety-nine participants (54 females) were recruited between August 2019 and June 2023. The flow of participants and loss to follow-up through the study is shown in Figure 1.

Characteristics of the participants

Descriptive statistics of participants in each intervention group are presented in Table 1. The three groups were well balanced at baseline. Notably, the baseline headache frequency was 6 days/month in all three groups; therefore, the smallest worthwhile effect threshold for headache frequency (ie, nominated as a 50% reduction in headache frequency) became 3 days/month.

Effects of the interventions

Primary outcome

Group means and standard deviations for the primary outcome measure – headache frequency – at all time points are presented in Table 2. Between-group differences in change with 95% CIs are presented in Table 3.

At 4 weeks, the mean difference in headache frequency achieved the smallest worthwhile effect threshold for MMT combined with exercise compared with either of the other groups, but the 95% CIs spanned that threshold (Table 3). Therefore, the results at 4 weeks indicated a benefit but the clinical relevance of the benefit was uncertain. At 13 and 26 weeks, these mean differences and 95% CIs achieved or exceeded the smallest worthwhile effect, confirming worthwhile benefits. Differences were negligible between sham MMT plus exercise and exercise alone at all time points.

Secondary outcomes

Headache parameters: Group means and standard deviations for headache intensity, headache duration, headache medication and headache disability at all time points are presented in Table 2. Between-group differences in change with 95% CI are presented in Table 3.

For headache intensity, the smallest worthwhile effect threshold was nominated as 1 point. When MMT plus exercise was compared

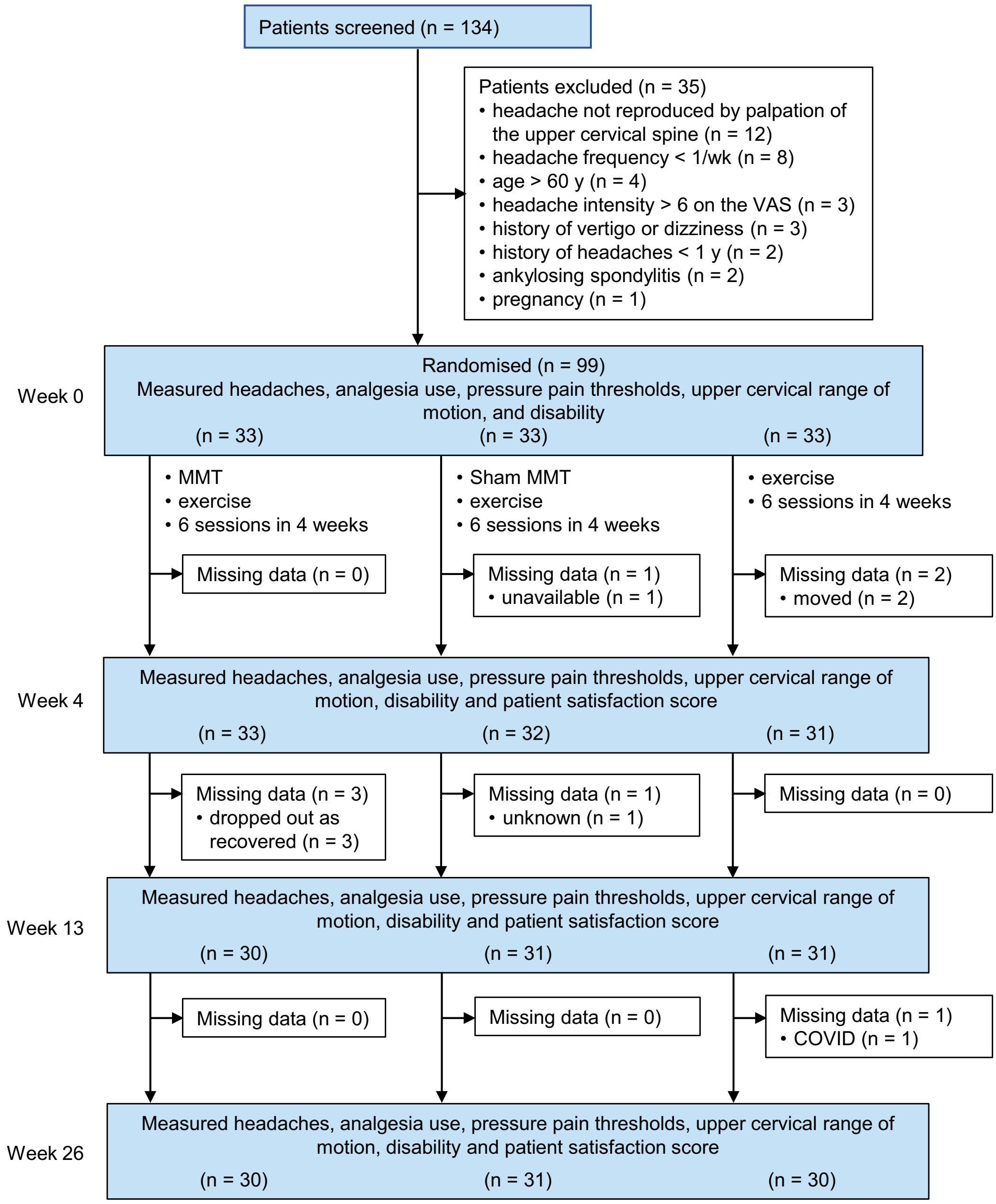

Figure 1. Design and flow of participants through the trial.

with the other two groups at 4 weeks, the mean between-group differences were below the smallest worthwhile effect and the 95% CIs spanned from close to no effect to beyond the smallest worthwhile effect threshold (Table 3), indicating a benefit, but making it uncertain whether the benefit was large enough to be clinically worthwhile. When MMT plus exercise was compared with the other two groups at 13 weeks, the mean between-group differences exceeded the smallest worthwhile effect and the 95% CIs spanned from around the smallest worthwhile effect threshold to a clearly worthwhile effect (Table 3), suggesting a worthwhile benefit. When MMT plus exercise was compared with the other two groups at 26 weeks, the mean between-group differences and the 95% CI both exceeded the smallest worthwhile effect (Table 3), clearly indicating a clinically worthwhile effect. Differences were negligible between sham MMT plus exercise and exercise alone at all time points.

For headache duration, the smallest worthwhile effect threshold was nominated as 7 hours/week. At 4 and 13 weeks, all between-group differences were negligible (ie, no confidence interval exceeded the smallest worthwhile effect). When MMT plus exercise was compared with the other two groups at 26 weeks, the mean between-group difference was around the smallest worthwhile effect and the 95% CIs spanned it (Table 3), indicating a benefit of uncertain clinical relevance. Differences were negligible between sham MMT plus exercise and exercise alone at all time points.

For headache medication intake, the smallest worthwhile effect threshold was nominated as 2.5 tablets/week. This effect was not achieved at any time point between any groups within any confidence interval, so all effects were negligible.

For headache disability, the smallest worthwhile effect threshold was nominated as 4.5 points. When MMT plus exercise was compared with the other two groups at 4 weeks, the mean between-group differences were marginally less than the smallest worthwhile effect and the 95% CIs spanned it fairly tightly (Table 3), indicating a benefit with a magnitude of uncertain clinical relevance. At later follow-up points, both the mean estimates and the confidence intervals exceeded the threshold, indicating clinically worthwhile benefits. Differences were negligible between sham MMT plus exercise and exercise alone at all time points.

Satisfaction: Group means and standard deviations for satisfaction are presented in Table 4. Between-group differences with 95% CIs are

Table 1
Baseline characteristics of participants.

Characteristic	MMT + ex (n = 33)	Sham MMT + ex (n = 33)	Ex (n = 33)
Age *(yr)*, mean (SD)	40 (11)	40 (13)	40 (11)
Sex, n (%)			
female	17 (52)	16 (48)	21 (64)
male	16 (48)	17 (52)	12 (36)
Height *(cm)*, mean (SD)	161 (7)	158 (9)	161 (8)
Weight *(kg)*, mean (SD)	67 (9)	65 (9)	68 (9)
Duration of CGH *(yr)*, mean (SD)	6 (4)	6 (2)	5 (1)
Occupation (n)			
clerk	3	5	3
driver	2	3	2
farmer	3	3	4
housewife	6	8	8
nurse	2	2	3
office worker	10	5	6
self-employed	4	3	0
student	1	1	2
teacher	2	2	3
wardsperson	0	1	2

CGH = cervicogenic headache, ex = exercise, MMT = Mulligan manual therapy.

presented in Table 5. Regarding the patient satisfaction score, the smallest worthwhile effect threshold nominated by patients in this study was 40%. When MMT plus exercise was compared with the other two groups at 4 weeks, the mean estimates just exceeded that threshold and the confidence intervals spanned it. At 13 and 26 weeks, the mean estimates and the confidence intervals all exceeded the threshold, indicating worthwhile benefits. Differences were negligible between sham MMT plus exercise and exercise alone at all time points.

Flexion rotation test: Group medians and interquartile ranges for the flexion rotation test at all time points are presented in Table 6. Median between-group differences with 95% CI are presented in Table 7. In a randomised trial,[29] upper cervical flexion-rotation ROM showed improvement concurrently with changes in headache following an MMT intervention and 10 deg was nominated as a large effect in that study. The smallest worthwhile effect for this outcome was therefore nominated as 5 deg.[44] When MMT plus exercise was compared with the other two groups at all post-randomisation time points, the confidence intervals exceeded that threshold (Table 7), indicating sustained worthwhile benefits. Differences were negligible between sham MMT plus exercise and exercise alone at all time points.

Pressure pain thresholds: Smallest worthwhile effects for pressure-pain thresholds were not nominated because they are more a mechanistic than clinical outcome. The largest effects were noted at the tibialis anterior, where MMT plus exercise exceeded the other two groups, with sustained effects across all time points. The zygapophyseal and suboccipital regions also indicated beneficial effects from MMT plus exercise over the other two interventions; these effects were small at week 4, increasing to moderate at week 26. Differences were very close to zero between sham MMT plus exercise and exercise alone at all time points for these three regions. The upper trapezius region had effects very close to zero between all groups at all time points.

Individual participant data are presented in Table 8. See the eAddenda for Table 8.

Assessment of the success of blinding

Responses to a blinding assessment questionnaire revealed that blinding was successful. A belief that manual therapy was not received was reported by 21 participants in the MMT with exercise group and 20 participants in the sham MMT with exercise group.

Discussion

This study reports the results of a randomised clinical trial conducted to evaluate the effects of MMT plus exercise compared with sham MMT plus exercise or exercise alone in the physiotherapy management of CGH. We were particularly interested in the effects on headache frequency, which showed immediate benefit favouring the MMT plus exercise group compared with the exercise alone group at the end of the 4-week intervention period. These effects increased in magnitude to become clinically worthwhile benefits at weeks 13 and 26. This suggests that most people with CGH are likely to find adding MMT to exercise worthwhile based on the reduction in headache frequency alone. Patients should also be advised that other outcomes showed similar benefits. For example, headache disability and upper cervical flexion-rotation ROM also showed clinically worthwhile benefits at 4, 13 and 26 weeks, whilst headache intensity and patient satisfaction showed worthwhile benefits at 13 and 26 weeks. Other benefits of uncertain clinical relevance were noted (eg, headache duration).

A very consistent finding among the results was that the difference between the MMT plus exercise group and the sham MMT plus exercise group was very similar to the difference between the MMT plus exercise group and the exercise only group. This indicates that the better results observed in the MMT plus exercise group over these other two groups were not due to placebo, especially given the success of blinding in the study.

Some of the thresholds for smallest worthwhile effect used in this study could only be nominated based on the clinical experience of medical practitioners having expertise in the management of headache; however, the fact that worthwhile benefits were identified on many outcomes suggests that adding MMT to exercise will be considered worthwhile by most people with CGH. This is especially so given that no adverse events were reported and that adding the MMT incurs relatively little additional treatment time.

Reduction in headache could be also linked with an indirect hypoalgesic effect of manual therapy and exercise causing improvement in cervical musculoskeletal dysfunction and consequent reduction in nociceptive input. Improvements were observed in pressure pain threshold at local sites and a remote area, which suggests that the mechanisms underlying the hypoalgesic effects of MMT were similar to other manual therapy techniques encompassing peripheral and central mechanisms.[45–54] However, further interpretation is limited because we did not nominate the smallest worthwhile effect for the pressure pain threshold.

Upper cervical spine ROM increased following MMT and was associated with a reduction in headache. One explanation for this could be that the restoration of normal movement at the C1–C2 motion segment decreased the strain on articular and periarticular mechanoreceptors and therefore reduced nociceptive signalling and provided new afferent input to the central nervous system by increasing non-nociceptive signalling.[33,52] Changes in the mechanoreceptor input have the potential to change local muscle activity, thus optimising muscle control in the upper cervical spine.

The pragmatic approach used in this study is replicable in real-life routine clinical practice, and along with successful blinding during

Table 2
Mean (SD) of headache outcomes for each group at each time point.

Outcome	Groups											
	Week 0			Week 4			Week 13			Week 26		
	MMT + ex (n = 33)	Sham MMT + ex (n = 33)	Ex (n = 33)	MMT + ex (n = 33)	Sham MMT + ex (n = 32)	Ex (n = 31)	MMT + ex (n = 30)	Sham MMT + ex (n = 31)	Ex (n = 31)	MMT + ex (n = 30)	Sham MMT + ex (n = 31)	Ex (n = 30)
Headache frequency *(d/mo)*	6 (1)	6 (1)	6 (1)	3 (1)	5 (1)	5 (1)	2 (1)	5 (1)	5 (1)	1 (1)	4 (1)	4 (1)
Headache intensity *(0 to 10)*	6.8 (0.5)	6.6 (0.4)	6.9 (0.6)	4.9 (1.1)	5.4 (0.9)	5.5 (0.9)	3.1 (1.0)	4.5 (0.9)	4.5 (0.9)	2.0 (1.0)	4.2 (0.9)	4.2 (0.8)
Headache duration *(hr/wk)*	15 (8)	14 (8)	16 (9)	10 (6)	12 (6)	13 (6)	7 (5)	10 (6)	11 (5)	4 (4)	9 (5)	10 (4)
Headache medication *(tablets/wk)*	3 (1)	3 (1)	3 (1)	1 (1)	3 (1)	2 (1)	1 (1)	2 (1)	2 (1)	1 (1)	2 (1)	2 (1)
Headache disability *(0 to 45)*	29 (3)	28 (4)	28 (3)	20 (3)	23 (4)	24 (3)	16 (3)	22 (5)	21 (3)	13 (3)	20 (4)	20 (3)

MMT = Mulligan manual therapy, ex = exercise.
Headache disability is measured using the Headache Activities of Daily Living index.

Table 3
Mean between-group difference (95% CI) in change in headache outcomes.

Outcome	Groups								
	Week 4 minus Week 0			Week 13 minus Week 0			Week 26 minus Week 0		
	MMT + ex minus Sham MMT + ex	MMT + ex minus ex	Sham MMT + ex minus ex	MMT + ex minus Sham MMT + ex	MMT + ex minus ex	Sham MMT + ex minus ex	MMT + ex minus Sham MMT + ex	MMT + ex minus ex	Sham MMT + ex minus ex
Headache frequency *(d/mo)*	−3 (−3 to −2)	−2 (−3 to −2)	−0 (0 to 1)	−3 (−4 to −3)	−3 (−4 to −3)	0 (−1 to 1)	−4 (−4 to −3)	−4 (−4 to −3)	0 (−1 to 1)
Headache intensity *(0 to 10)*	−0.8 (−1.2 to −0.4)	−0.7 (−1.1 to −0.2)	0.1 (−0.3 to 0.6)	−1.6 (−2.1 to −1.2)	−1.3 (−1.8 to −0.8)	0.3 (−0.2 to 0.7)	−2.4 (−2.9 to −1.7)	−2.1 (−2.7 to −1.6)	0.3 (−0.2 to 0.7)
Headache duration *(hr/wk)*	−3 (−5 to −2)	−3 (−4 to −1)	1 (−1 to 2)	−4 (−6 to −2)	−4 (−6 to −1)	0 (−1 to 2)	−6 (−9 to −4)	−5 (−8 to −3)	1 (−1 to 3)
Headache medication *(tablets/wk)*	−1 (−2 to −1)	−1 (−2 to −1)	0 (0 to 0)	−1 (−2 to −1)	−1 (−2 to −1)	0 (−1 to 1)	−1 (−2 to −1)	−1 (−2 to −1)	0 (−1 to 0)
Headache disability *(0 to 45)*	−4 (−5 to −3)	−4 (−5 to −3)	0 (−1 to 1)	−7 (−8 to −6)	−6 (−7 to −5)	1 (−1 to 2)	−8 (−10 to −7)	−7 (−9 to −6)	1 (−1 to 3)

MMT = Mulligan manual therapy, ex = exercise.
Headache disability is measured using the Headache Activities of Daily Living index.

Table 4
Mean (SD) satisfaction for each group at each time point.

Outcome	Groups								
	Week 4			Week 13			Week 26		
	MMT + ex (n = 33)	Sham MMT + ex (n = 32)	Ex (n = 31)	MMT + ex (n = 30)	Sham MMT + ex (n = 31)	Ex (n = 31)	MMT + ex (n = 30)	Sham MMT + ex (n = 31)	Ex (n = 30)
Patient satisfaction score *(0 to 100)*	71 (19)	29 (16)	30 (16)	79 (18)	25 (12)	26 (13)	84 (15)	25 (11)	26 (13)

MMT = Mulligan manual therapy, ex = exercise.

Table 5
Mean difference (95% CI) in satisfaction between groups.

Outcome	Groups								
	Week 4			Week 13			Week 26		
	MMT + ex minus Sham MMT + ex	MMT + ex minus ex	Sham MMT + ex minus ex	MMT + ex minus Sham MMT + ex	MMT + ex minus ex	Sham MMT + ex minus ex	MMT + ex minus Sham MMT + ex	MMT + ex minus ex	Sham MMT + ex minus ex
Patient satisfaction score *(0 to 100)*	42 (34 to 51)	42 (33 to 50)	–1 (–9 to 7)	54 (46 to 62)	53 (45 to 61)	–1 (–7 to 5)	59 (52 to 66)	58 (51 to 65)	–1 (–7 to 5)

MMT = Mulligan manual therapy, ex = exercise.

Table 6
Median (IQR) of other outcomes for each group at each time point.

Outcome	Groups											
	Week 0			Week 4			Week 13			Week 26		
	MMT + ex (n = 33)	Sham MMT + ex (n = 33)	Ex (n = 33)	MMT + ex (n = 33)	Sham MMT + ex (n = 32)	Ex (n = 31)	MMT + ex (n = 30)	Sham MMT + ex (n = 31)	Ex (n = 31)	MMT + ex (n = 30)	Sham MMT + ex (n = 31)	Ex (n = 30)
Flexion rotation test *(deg)*	26 (24 to 29)	28 (25 to 29)	27 (25 to 29)	41 (38 to 43)	30 (28 to 34)	30 (29 to 22)	43 (41 to 44)	32 (30 to 35)	32 (31 to 34)	44 (43 to 45)	33 (31 to 35)	33 (31 to 35)
PPT suboccipital muscle area *(kg/cm^2)*	1.2 (0.8 to 1.4)	1.3 (1.1 to 1.5)	1.2 (0.8 to 1.4)	1.4 (1.1 to 1.5)	1.3 (1.1 to 1.6)	1.3 (0.9 to 1.5)	1.8 (1.6 to 2.0)	1.4 (1.2 to 1.6)	1.3 (0.9 to 1.5)	1.9 (1.6 to 2.1)	1.4 (1.2 to 1.7)	1.3 (1.0 to 1.6)
PPT zygapophyseal joint area *(kg/cm^2)*	1.2 (1.1 to 1.4)	1.2 (1.0 to 1.6)	1.3 (0.9 to 1.5)	1.5 (1.3 to 1.7)	1.2 (1.0 to 1.6)	1.3 (1.0 to 1.5)	1.5 (1.3 to 1.7)	1.3 (1.0 to 1.6)	1.3 (1.0 to 1.5)	1.7 (1.4 to 1.8)	1.3 (1.1 to 1.7)	1.4 (1.1 to 1.6)
PPT upper trapezius muscle area *(kg/cm^2)*	2.4 (2.0 to 2.6)	2.4 (2.0 to 2.7)	2.5 (2.2 to 2.6)	2.4 (2.0 to 2.6)	2.4 (2.0 to 2.7)	2.5 (2.2 to 2.7)	2.5 (2.0 to 2.7)	2.4 (2.0 to 2.7)	2.5 (2.2 to 2.7)	2.5 (2.2 to 2.8)	2.4 (2.0 to 2.8)	2.6 (2.2 to 2.7)
PPT tibialis anterior muscle area *(kg/cm^2)*	4.1 (3.3 to 5.1)	4.0 (3.5 to 4.6)	4.1 (3.3 to 5.1)	6.4 (5.6 to 7.2)	4.1 (3.6 to 4.7)	4.1 (3.6 to 4.7)	6.4 (5.7 to 7.2)	4.1 (3.6 to 4.7)	4.1 (3.6 to 4.7)	6.9 (6.2 to 7.7)	4.1 (3.7 to 4.7)	4.1 (3.6 to 4.8)

MMT = Mulligan manual therapy, ex = exercise, PPT = pressure pain threshold.

Table 7
Median between-group difference (95% CI) in change in other outcomes.

Outcome	Groups								
	Week 4			Week 13			Week 26		
	MMT + ex minus Sham MMT + ex	MMT + ex minus ex	Sham MMT + ex minus ex	MMT + ex minus Sham MMT + ex	MMT + ex minus ex	Sham MMT + ex minus ex	MMT + ex minus Sham MMT + ex	MMT + ex minus ex	Sham MMT + ex minus ex
Flexion rotation test *(deg)*	11 (10 to 13)	11 (9 to 12)	0 (–2 to 1)	12 (10 to 13)	11 (10 to 13)	–1 (–2 to 1)	12 (10 to 14)	12 (10 to 13)	0 (–2 to 1)
PPT suboccipital muscle area *(kg/cm^2)*	0.1 (0.1 to 0.2)	0.1 (0.1 to 0.2)	0.0 (0.0 to 0.0)	0.6 (0.4 to 0.7)	0.6 (0.4 to 0.7)	0.0 (0.0 to 0.1)	0.6 (0.4 to 0.7)	0.6 (0.4 to 0.8)	0.0 (0.0 to 0.1)
PPT zygapophyseal joint area *(kg/cm^2)*	0.2 (0.1 to 0.3)	0.2 (0.1 to 0.2)	0.0 (0.0 to 0.0)	0.3 (0.2 to 0.3)	0.3 (0.2 to 0.3)	0.0 (–0.1 to 0.0)	0.3 (0.3 to 0.4)	0.3 (0.2 to 0.4)	0.0 (–0.1 to 0.0)
PPT upper trapezius muscle area *(kg/cm^2)*	0.0 (0.0 to 0.1)	0.0 (0.0 to 0.0)	0.0 (0.0 to 0.0)	0.1 (0.0 to 0.1)	0.0 (0.0 to 0.1)	0.0 (–0.1 to 0.0)	0.1 (0.0 to 0.1)	0.1 (0.0 to 0.1)	0.0 (0.0 to 0.0)
PPT tibialis anterior muscle area *(kg/cm^2)*	2.4 (2.1 to 2.5)	2.4 (2.1 to 2.5)	0.0 (0.0 to 0.0)	1.7 (1.0 to 2.2)	1.7 (1.0 to 2.2)	0.0 (0.0 to 0.0)	2.2 (1.8 to 2.9)	2.1 (1.7 to 2.8)	0.0 (0.0 to 0.1)

MMT = Mulligan manual therapy, ex = exercise, PPT = pressure pain threshold.

the intervention period, increases the external validity of the study findings.[55] The participants maintained excellent adherence to the intervention as well as follow-up sessions. An important aspect of this study is the safety of MMT, which was confirmed by the lack of adverse events. The overlapping nature of symptoms is common in different forms of headache, which may have prevented accurate categorisation of CGH.[56] Diagnosis of CGH was based on clinical criteria and anaesthetic blockades were not available.[4] Incorrect classification may have reduced the effect of MMT intervention. The possibility of patients seeking other contemporary treatment options could not be excluded, although none were reported. Finally, the Headache Activities of Daily Living Index questionnaire has not undergone extensive clinimetric validation.

The results of this study provide evidence for the effectiveness of the addition of MMT to exercise for the long-term management of CGH in terms of reducing headache frequency, intensity and disability, but not for reducing headache duration. Similarly, MMT and exercise were effective in the long-term management of cervical musculoskeletal impairments including upper cervical rotation ROM and pressure pain threshold assessed at the suboccipital area, which are potentially important factors in the generation of headache. Furthermore, this study confirmed that these benefits were not due to a placebo response.

What was already known on this topic: Cervicogenic headache is a non-throbbing, unilateral, side-locked headache that is secondary to cervical musculoskeletal dysfunction. It is often aggravated by sustained neck postures, repeated neck movements or physical activity. There is preliminary evidence that individual Mulligan manual therapy techniques have effects on aspects of cervicogenic headache, but the long-term effects of a pragmatic Mulligan manual therapy treatment approach have not been reported.
What this study adds: In people with cervicogenic headache, adding Mulligan manual therapy to exercise improves to a clinically worthwhile extent: headache frequency, intensity and duration; headache-related disability; upper cervical rotation; and patient satisfaction. These benefits were sustained to at least 26 weeks. These benefits were not due to placebo effects.

Footnotes: [a] SPSS software, version 26.0, Chicago, SPSS, USA.

eAddenda: Table 8 can be found online at https://doi.org/10.1016/j.jphys.2024.06.002

Ethics approval: The trial was approved by the local ethics committee: SKNCOPT/IEC/2019/208 and IRB-SIOR/Agenda 049. All participants gave written informed consent before data collection began.

Competing interests: Kiran Satpute is a member of the Mulligan Concept Teacher Association and gains a teaching fee when running Mulligan Concept courses.

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